3D Reconstruction of Biomedical Tissues from Sparse 2D CT/MRI Slices Using Conditional Diffusion Models

Abstract—The reconstruction of highly detailed, isotropic three-dimensional (3D) anatomical volumes from sparse, two-dimensional (2D) biomedical imaging modalities represents a foundational challenge in computational pathology and clinical radiology. Traditional constraints in Computed Tomography (CT), such as radiation dose limits, and in Magnetic Resonance Imaging (MRI), such as extended acquisition times, inherently force protocols to rely on sparse-view projections, limited-angle captures, or thick-slice acquisitions. The resulting data is characterized by severe anisotropy, presenting high in-plane resolution but critically degraded through-plane resolution. While classical interpolation and early generative adversarial networks (GANs) have been utilized to hallucinate missing anatomical data, these approaches suffer from blurring, mode collapse, and the generation of physically impossible structural artifacts. Over the past three years (2023–2026), conditional diffusion models have emerged as a paradigm-shifting solution. By reframing biomedical image reconstruction as a conditional generative modeling problem, these frameworks decouple the learning of high-fidelity anatomical priors from the specific physics of measurement operators. This comprehensive literature review examines a minimum of 15 recent, state-of-the-art frameworks that leverage conditional diffusion models for the 3D reconstruction of tissues from sparse 2D scans. The review systematically categorizes algorithmic advancements into four critical domains: (1) mechanisms for enforcing inter-slice volumetric coherence and mitigating stochastic inconsistencies; (2) the integration of explicit physical forward models into Plug-and-Play (PnP) solvers to ensure data fidelity; (3) innovations in cross-modality domain translation and zero-shot contrast adaptation; and (4) architectural optimizations designed to reduce inference latency for clinical deployment. Ultimately, this analysis demonstrates that the synthesis of diffusion priors with physics-informed operators provides an unprecedented capability to recover diagnostic-quality, isotropic 3D volumes from highly impoverished clinical data.
Index Terms—Biomedical Image Reconstruction, Conditional Diffusion Models, Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Sparse-View Imaging, Super-Resolution, Volumetric Coherence.

I. Introduction

The necessity for high-resolution, isotropic three-dimensional (3D) medical imaging is paramount in modern clinical workflows, underpinning accurate diagnostic evaluation, surgical planning, and longitudinal disease monitoring. Imaging modalities such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) capture intricate cross-sectional representations of internal anatomy[1]. However, the acquisition of fully dense, isotropic 3D volumes is frequently precluded by fundamental physical, biological, and logistical limitations[2].

In the domain of CT imaging, minimizing the ionizing radiation dose delivered to the patient is a strict clinical imperative. This dose-reduction requirement forces technicians to acquire scans using sparse-view projections or limited-angle trajectories, which subsequently violate the Nyquist-Shannon sampling theorem[1]. Reconstructing these sparse projections using standard analytical methods, such as Filtered Back Projection (FBP), inevitably yields severe streaking artifacts and significant signal degradation[2]. Conversely, in MRI, achieving high spatial resolution necessitates extended scan durations, which inherently increases patient discomfort, reduces clinical throughput, and elevates the probability of motion-induced artifacts[2]. To circumvent these issues, standard MRI protocols frequently acquire thick 2D slices with substantial spatial gaps between them, resulting in highly anisotropic volumetric datasets where the in-plane ($x - y$) resolution far exceeds the through-plane (z-axis) resolution[4].

The classical approach to resolving this anisotropy relies on mathematical interpolation techniques, such as linear, cubic, spline, or nearest-neighbor interpolation[1]. These techniques, which calculate intermediate voxel intensities based on local polynomial functions, lack a fundamental understanding of human anatomy. Consequently, they fail to accurately reconstruct high-frequency details, often blurring fine tissue boundaries and obscuring micro-lesions[1]. Subsequently, Compressed Sensing (CS) techniques were widely adopted, leveraging the inherent sparsity of medical images within specific transform domains (e.g., wavelet domains or total variation penalties) to reconstruct missing data[2]. While mathematically robust, CS techniques require exhaustive, iterative non-linear optimization algorithms that struggle to adequately model complex, heterogeneous biological textures[2].

The advent of deep learning catalyzed a profound shift in medical image super-resolution and reconstruction. Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs) superseded CS as the optimal mechanism for mapping sparse inputs to dense targets[9]. However, the application of GANs in safety-critical medical contexts is inherently problematic. GANs optimize for perceptual realism rather than strict data fidelity, rendering them susceptible to mode collapse and "hallucination"—the generation of photorealistic but anatomically false structures that can lead to catastrophic misdiagnoses[3].

In the contemporary literature landscape (2023–2026), conditional diffusion models—comprising Denoising Diffusion Probabilistic Models (DDPMs), Score-Based Stochastic Differential Equations (SDEs), and Brownian Bridge Diffusion Models (BBDMs)—have established a new gold standard for medical inverse problems[10]. These generative models operate by systematically corrupting an image with Gaussian noise and subsequently learning to reverse this process, conditioned on available sparse clinical data[9]. Diffusion models offer superior distribution matching, greater training stability, and an unprecedented capacity to quantify predictive uncertainty compared to their adversarial predecessors[2]. This review critically evaluates over 15 recent papers and algorithmic frameworks that utilize conditional diffusion models for the 3D reconstruction of biomedical tissues from sparse 2D slices. The analysis dissects the evolution of these models, focusing on the critical challenges of maintaining volumetric coherence across slices, enforcing physics-informed data consistency, enabling cross-modal zero-shot translation, and achieving the inference speeds requisite for

real-world clinical deployment.

II. Mathematical and Theoretical Foundations of Conditional Diffusion

To contextualize the advancements in recent literature, it is necessary to establish the theoretical mechanics of diffusion models as applied to medical image reconstruction. At their core, diffusion models are generative frameworks inspired by non-equilibrium thermodynamics, defined by two opposing Markov chains: a forward diffusion process and a reverse denoising process[9].

A. The Forward and Reverse Processes

The forward process, denoted as $q(x_t|x_{t-1})$, systematically degrades a pristine medical image slice x_0 by injecting isotropic Gaussian noise over a discrete sequence of timesteps $t \in [1, T]$ [9]. The magnitude of the noise variance added at each specific timestep is governed by a predetermined variance schedule β_t . Over a sufficiently large number of steps T , the original structural information is entirely obliterated, and the final state x_T converges to a standard normal distribution $\mathcal{N}(0, I)$ [10].

The fundamental power of the model resides in the reverse process, $p_\theta(x_{t-1}|x_t)$, where a deep neural network—typically a modified U-Net architecture featuring extensive spatial self-attention and cross-attention mechanisms—is optimized to learn the inverse trajectory[1]. By predicting and subtracting the specific noise component at each timestep, the model iteratively refines a sample drawn from pure noise back into a highly structured, high-fidelity anatomical image x_0 [9].

B. Conditioning for Medical Inverse Problems

In the context of biomedical tissue reconstruction, unconditional generation is clinically useless; the generated tissue must belong explicitly to the specific patient being examined. Therefore, the diffusion process must be strictly conditional[1]. Given an acquired sparse dataset y (e.g., a limited number of 2D slices, or an undersampled k-space matrix), the objective is to model the conditional posterior distribution $p(x|y)$ [11].

The literature demonstrates several sophisticated conditioning modalities. The most direct approach is spatial conditioning, where the known adjacent 2D slices are concatenated with the noisy target slice along the channel dimension at the input layer of the U-Net[1]. This provides the network with explicit boundary constraints, forcing the denoised slice to geographically align with the surrounding tissue[15]. More advanced techniques utilize transform-domain

conditioning, embedding frequency-specific prompts or wavelet decompositions into the network to separate the generation of low-frequency macro-structures from high-frequency micro-textures[8]. The integration of these conditions allows diffusion models to act as robust priors for a vast array of medical inverse problems, shifting the paradigm from purely data-driven interpolation to physics-guided generative reconstruction[3].

III. Algorithmic Advancements in Volumetric Coherence

A pervasive obstacle identified across the recent literature is the strict dimensional dichotomy between the nature of human anatomy and the computational limits of deep learning hardware. Anatomical tissues are continuous 3D structures, yet training fully volumetric 3D diffusion models requires immense Graphics Processing Unit (GPU) memory capacities and massive cohorts of perfectly isotropic 3D scans, which are rarely available[17]. To bypass this bottleneck, the prevailing strategy is to train 2D diffusion models and apply them iteratively, slice-by-slice along the z-axis, to construct a 3D volume[17].

However, this naive stacking of 2D generated slices introduces a critical vulnerability: the intrinsic stochasticity of the reverse diffusion sampling process. Because the reverse process for each missing slice begins from an independent, randomly sampled Gaussian noise vector x_T, adjacent slices follow entirely distinct, uncorrelated generative trajectories[17]. This lack of correlation manifests as severe slice inconsistency in the final 3D volume, characterized by sudden, unnatural fluctuations in tissue brightness, contrast, and morphological topology[18]. Over the past three years, researchers have developed highly innovative mechanisms to enforce inter-slice continuity without defaulting to the computational burden of true 3D models.

A. Stochastic Trajectory Alignment and Spherical Interpolation

Addressing the root cause of these volumetric discontinuities, Du et al. (2026) introduced Inter-Slice Consistent Stochasticity (ISCS), a mathematical strategy designed to constrain the latent noise manifold during sampling[20]. The authors hypothesized that if two anatomical slices are structurally adjacent and continuous in physical space, their corresponding starting points in the pure noise space x_T must not be mathematically independent. Instead of drawing entirely separate Gaussian vectors for adjacent slices, ISCS utilizes Spherical Linear Interpolation (Slerp) to explicitly correlate the stochastic noise structures across the z-axis[17]. By interpolating the noise vectors along the surface of a high-dimensional hypersphere, Slerp ensures that the unit variance expected by the diffusion prior is perfectly maintained, avoiding the signal attenuation that would occur with standard linear interpolation[17]. This alignment forces the parallel reverse sampling trajectories of adjacent slices to remain synchronized throughout all T timesteps. The profound advantage of ISCS is that it is entirely

"plug-and-play"; it requires zero additional loss terms, no extra optimization steps, and fundamentally no retraining of the underlying 2D diffusion model[17]. Demonstrated on sparse-view CT (SVCT), limited-angle CT (LACT), and MRI z-axis super-resolution tasks, ISCS dramatically reduced inter-slice artifacts, outperforming traditional total variation (TV) regularizations that typically result in over-smoothed, blurred volumes[17].

B. Brownian Bridge Trajectories and Style Key Conditioning

In 2024, Choo et al. presented a highly effective framework for Slice-Consistent 3D Volumetric Brain CT-to-MRI Translation utilizing a 2D Brownian Bridge Diffusion Model (BBDM)[12]. Standard DDPMs rely on a Gaussian prior, which is optimal for unconditional generation but sub-optimal for domain translation. A BBDM, conversely, constructs a diffusion bridge directly between two distinct empirical distributions (e.g., from the source CT domain directly to the target MRI domain), facilitating highly stable image-to-image mappings[12].

To resolve the 3D inconsistency inherent to 2D models, Choo et al. engineered two simultaneous interventions: Style Key Conditioning (SKC) and Inter-Slice Trajectory Alignment (ISTA)[18]. SKC targets the photometric inconsistencies (brightness and contrast shifts) by injecting a target histogram-based style key directly into the U-Net architecture, forcing all independently generated slices to adhere to a unified, global imaging style[12]. Concurrently, ISTA addresses the geometric and morphological discontinuities. Operating within a predictor-corrector sampling paradigm, ISTA aggregates the outputs of multiple adjacent slices during the co-prediction phase[19]. It deterministically calculates an agreed-upon directional vector for the next diffusion timestep, mathematically interlocking the independent slices into a unified shape trajectory[18]. Validated on in-house datasets and the BraTS2023 benchmark, the combination of SKC and ISTA enabled this purely 2D diffusion framework to quantitatively and qualitatively surpass highly complex, computationally heavy 3D GAN baselines, achieving a Normalized Root Mean Square Error (NRMSE) of 0.0454 and a Peak Signal-to-Noise Ratio (PSNR) of 27.589[18].

C. Deformable Field Extraction and Optical Flow

An alternative approach to enforcing depth-wise coherence borrows fundamental principles from video frame interpolation. In 2026, Wang et al. proposed DF-DiffVSR, a Deformable Field-Driven Diffusion Model explicitly engineered for medical volume super-resolution[7]. A primary limitation of standard slice-wise synthesis models is their restricted spatial receptive field, which fails to capture global anatomical contexts spanning multiple slices[7]. DF-DiffVSR mitigates this through a Deformable Field Extraction (DFE) module, which integrates optical flow estimations directly into the diffusion architecture[7].

The DFE module actively learns the inter-slice motion and morphological displacement vectors between the known, thick superior and inferior slices. By establishing this deformable flow field, the network mathematically warps the contextual features from the adjacent bounding slices to perfectly guide the structural generation of the missing intermediate thin slice[7]. To further enhance global awareness, the architecture utilizes a Multiscale Large Kernel

Convolution (MLKC) module, employing striped convolutions with varying kernel sizes to aggressively expand the receptive field[7]. Evaluated on the RPLHR-CT and IXI-T2 datasets, DF-DiffVSR established state-of-the-art volumetric continuity, surpassing sub-optimal baseline methods by 0.732 dB in PSNR, demonstrating a profound capacity to recover fine-grained, continuous lesions[7].

D. State-Space Models and Boundary Prompts

The temporal dimension introduces further complexity, particularly in 4D dynamic MRI (e.g., cardiac or knee joint imaging), where the trade-off between spatial and temporal resolution requires extended scan times that invariably capture patient motion[24]. Classical registration-based temporal interpolation techniques suffer catastrophic failures during large-amplitude deformations[24]. In 2025, Zhou et al. introduced TSSC-Net, a temporal super-resolution and spatial consistency enhancement framework[24]. To resolve the extreme spatial inconsistencies caused by cross-slice misalignment during rapid motion, TSSC-Net abandons standard self-attention (which scales quadratically and struggles with massive 3D volumetric contexts) in favor of a tri-directional Mamba-based module[24]. Leveraging state-space models, the Mamba architecture efficiently processes immense, long-range contextual sequences, actively identifying and correcting volumetric cross-slice topological errors while achieving a 6x temporal super-resolution in a single step[24].

Topological fidelity is also the primary focus of Diff-UNet (Zhu et al., 2025) and SpineSynNet (2025). Diff-UNet addresses 3D medical image segmentation and slice synthesis by appending a dedicated boundary-prediction branch to the diffusion network[25]. Through a Multi-granularity Boundary Aggregation (MBA) module, explicit topological limits are injected into the diffusion denoising sequence, creating a rigid mathematical fence that prevents the stochastic hallucination of tissues beyond known anatomical margins[25]. SpineSynNet tackles anisotropic thick-slice spinal MR images by completely abandoning the standard axial-only generation pipeline[26]. Through a View-prompt Deformation Field Estimation Module (VDFEM), SpineSynNet evaluates the sparse volume simultaneously from the axial, coronal, and sagittal planes[26]. This multi-view prompting ensures that the hallucinated intermediate slices satisfy rigid geometric and anatomical constraints from all orthogonal directions simultaneously, vastly improving 3D realism[26].

IV. Physics-Guided Data Consistency and Inverse Problem Formulations

While generative realism is critical, biomedical 3D reconstruction is fundamentally an inverse problem governed by the equation $y = Ax + \epsilon$. Here, y represents the acquired sparse clinical measurements (e.g., an undersampled k-space matrix or a limited-angle sinogram), A is the deterministic physical forward measurement operator of the scanner, x is the true

underlying anatomical volume, and ϵ represents inherent system noise[9]. Generating visually pristine slices is a clinical hazard if the resulting 3D volume mathematically contradicts the raw measurements in y. Consequently, a major focus of recent literature is the strict integration of physical forward models into the diffusion sampling loop to ensure absolute data fidelity.

A. Dual-Coupled ADMM and Plug-and-Play Diffusion

Plug-and-Play (PnP) frameworks represent a powerful paradigm in computational imaging. They decouple the reconstruction process into two alternating steps: a data-consistency optimization step (enforcing fidelity to y) and a generative denoising step (enforcing adherence to the anatomical prior learned by the diffusion model)[28]. However, Du et al. (2026) identified a critical geometric flaw in the prevailing PnP solvers that utilize Proximal Gradient Descent or Half-Quadratic Splitting (HQS)[28]. These standard solvers act as memoryless operators; they update the volume estimate based solely on the instantaneous gradient of the current step[28]. Under heavy signal corruption—such as ultra-sparse CT views—this lack of historical memory leads to a non-vanishing steady-state bias. The final 3D reconstruction visually resembles anatomy but mathematically violates the raw physical measurements[28].

To rectify this, the researchers developed Dual-Coupled PnP Diffusion, completely replacing the naive proximal steps with the Alternating Direction Method of Multipliers (ADMM)[30]. The integration of ADMM restores the classical dual variable to the optimization loop, providing integral, historical feedback that theoretically guarantees asymptotic convergence to the exact physical data manifold[28].

Yet, this rigorous mathematical coupling introduced a severe secondary challenge. The accumulated dual residuals generated by ADMM exhibit spectrally colored, highly structured artifact patterns. Because standard diffusion priors are trained strictly to remove Additive White Gaussian Noise (AWGN), feeding these colored artifacts into the U-Net denoiser violated the core statistical assumptions of the model, triggering catastrophic generative hallucinations[28]. To bridge this critical gap, Du et al. engineered Spectral Homogenization (SH)[28]. SH operates as a frequency-domain adaptation mechanism, systematically modulating the structured dual residuals into statistically compliant pseudo-AWGN inputs[28]. This innovation successfully aligns the solver's rigorous physical optimization trajectory with the denoiser's valid statistical manifold, entirely resolving the bias-hallucination trade-off and achieving unprecedented fidelity in highly undersampled environments[28].

B. Uncertainty Quantification in Limited-Angle CT

In Limited-Angle Computed Tomography (LACT), physical obstructions or clinical restrictions prevent the X-ray source from completing a full 180-degree sweep[2]. The resulting data loss is not randomly distributed but localized to specific angular wedges, which traditionally causes massive structural distortions in the z-axis depth[2]. Addressing this, Liu et al. (2023) developed DOLCE, a Model-Based Probabilistic Diffusion Framework optimized explicitly for LACT[2].

DOLCE tightly integrates a conditionally-trained diffusion model with the explicit physical measurement models of the CT system[3]. By alternating between strict data-fidelity updates and the stochastic sampling updates of the diffusion model (conditioned on the transformed limited-angle sinogram), DOLCE synthesizes highly accurate anatomical structures in the obscured wedges[2]. Critically, because DOLCE leverages a probabilistic sampler rather than a deterministic regression network, running the algorithm multiple times on the exact same sparse sinogram yields a diverse set of plausible 3D reconstructions[2]. By calculating the pixel-wise variance across these multiple generated samples, DOLCE produces well-calibrated, highly accurate uncertainty maps[2]. These maps are vital for clinical safety, providing radiologists with a mathematically sound estimation of reconstruction confidence, highlighting precisely which regions of the hallucinated tissue are statistically reliable versus which regions require further clinical investigation[2].

C. Wavelet and Frequency-Domain Optimization

Reconstructing ultra-sparse-view CT poses distinct optimization challenges. Xu et al. (2024) noted that applying diffusion models directly in the raw sinogram or spatial image domains frequently leads to training instability, causing the models to prematurely converge on local minimal solutions due to the extreme dynamic range of CT Hounsfield units[8]. To ensure robust convergence, the authors proposed SWORD (Stage-by-stage Wavelet Optimization Refinement Diffusion), an innovative architecture that physically shifts the entire diffusion process into the wavelet domain[8].

The wavelet transform effectively disentangles the raw image contents into highly specific, multi-scale frequency bands[8]. SWORD executes reconstruction via a two-stage hierarchical generative process. First, the Wavelet-based Full-frequency Diffusion Model (WFDM) evaluates the low-frequency bands to aggressively establish the broad structural profile and macroscopic anatomical geometry from the sparse projections[8]. Subsequently, the Wavelet-based High-frequency Diffusion Model (WHDM) targets the high-frequency components, strictly refining directional textures, micro-lesions, and fine tissue boundaries without altering the macro-structure[8]. This frequency-guided decoupling vastly insulates the neural network against the intense streaking artifacts typical of sparse-view CT, allowing for highly stable training and superior structural recovery[8]. A parallel philosophy is evident in the work of Xie et al. (2024), who utilized a Prior Frequency Guided Diffusion Model (PFGDM) to systematically guide the restoration of lost high-frequency spatial frequencies in limited-angle Cone-Beam CT (CBCT) datasets used in radiotherapy[16].

V. Cross-Modality Translation and Domain Shift Adaptation

In numerous clinical oncology and neurology workflows, comprehensive diagnosis requires multimodal evaluation. A patient may present with a highly sparse, low-dose, thick-slice CT scan, yet the clinical protocol demands the soft-tissue contrast clarity only available in a dense 3D MRI volume[37]. This creates the need for cross-modality medical image-to-image translation.

Generating dense MRI volumes from sparse CT slices is immensely complex, demanding the algorithm to simultaneously perform geometric spatial interpolation (to fill the physical gaps) and non-linear photometric mapping (to translate Hounsfield units into MRI signal intensities)[37].

A. Retrieval-Augmented Generation (RAG) in Diffusion

Liu et al. (2026) addressed this dual challenge by introducing ReBrain, a revolutionary framework that integrates Retrieval-Augmented Generation (RAG) mechanisms into medical diffusion models[37]. Given an input 3D CT scan suffering from severely sparse slices, ReBrain first extracts deep semantic features and utilizes a fine-tuned retrieval model to search a massive, comprehensive prior database[37]. The system retrieves structurally and pathologically analogous high-resolution CT slices from historical patient archives.

These retrieved, dense anatomical slices serve as rigid external reference priors. ReBrain injects these reference priors into a 2D Brownian Bridge Diffusion Model (BBDM) via a dedicated ControlNet branch[37]. The ControlNet actively modulates the generative trajectory of the BBDM, ensuring that the hallucinated intermediate MRI slices adhere strictly to the complex human topologies present in the retrieved reference data[37]. In the rare scenario of a retrieval failure—such as a patient presenting with an unprecedented pathological anomaly not present in the database—ReBrain dynamically reverts to spherical linear interpolation to provide supplementary geometric guidance[37]. Extensively validated on the SynthRAD2023 and BraTS datasets, the integration of retrieval-augmented priors allowed ReBrain to achieve unparalleled state-of-the-art performance in cross-modal volume reconstruction under extremely sparse conditions[37].

B. Zero-Shot Contrast Transport for Low-Field MRI

A persistent obstacle to the global democratization of high-quality medical imaging is the prohibitive cost of high-field (HF) MRI scanners (e.g., 3.0 Tesla). Low-field (LF) MRI scanners (e.g., 0.5 Tesla) are vastly more accessible but inherently suffer from suppressed signal-to-noise ratios and severe field-dependent distortions[40]. Because the longitudinal (T_1) and transverse (T_2) relaxation times of biological tissues shift dynamically relative to the main magnetic field strength, LF scans exhibit a non-linear compression of the image's intensity distribution[40]. This contrast distortion often mutes critical diagnostic boundaries, such as the differentiation between gray and white matter in neurology[40]. Reconstructing HF-quality volumes from LF slices represents a blind inverse problem, heavily hindered by the physical impossibility of acquiring perfectly paired, simultaneous LF and HF scans of the same patient[41]. To solve this, Liu et al. (2026) developed DACT (Diffusion-Based Adaptive Contrast Transport), a pioneering zero-shot framework capable of rectifying LF distortions entirely without paired supervision[41]. Recognizing that the standard linear degradation assumptions used by previous models fundamentally fail to capture the complex physics of magnetic relaxation, DACT synergizes a pre-trained HF anatomical diffusion prior with a highly advanced, physically-informed adaptive forward model[40].

The core innovation of DACT is the integration of a differentiable Sinkhorn Optimal Transport

module directly into the reverse diffusion sequence[40]. Standard histogram matching is discrete and non-differentiable, rendering it useless for gradient-based deep learning optimization[40]. By applying the Sinkhorn algorithm (which introduces entropic regularization to the optimal transport problem), DACT achieves monotonic, mathematically differentiable histogram matching[40]. The module computes Kernel Density Estimations (KDE) to generate soft histograms of both the observed LF image and the current HF estimate, subsequently calculating the optimal transport plan to align their intensity distributions[40]. This allows the diffusion network to dynamically learn and correct the intractable, non-linear contrast mapping entirely on-the-fly, guided solely by the marginal distributions of the two domains. The result is the successful restoration of authentic HF tissue contrast and structural detail from heavily distorted LF scans[40].

Furthermore, to ensure models generalize safely across diverse hospital networks (where scanner hardware and software protocols vary wildly), Güngör et al. (2023) developed AdaDiff, an adaptive diffusion prior designed for accelerated MRI reconstruction[43]. AdaDiff actively monitors the incoming k-space measurements and dynamically fine-tunes its generative sampling trajectory to accommodate out-of-distribution data[43]. This adaptive tracking prevents the diffusion prior from forcing the reconstructed volume into a latent distribution that violates the specific hardware characteristics of the active clinical site, significantly improving the robustness of the model against domain shifts[43].

VI. Algorithmic Efficiency and Clinical Feasibility

Despite the overwhelming evidence of their superior generative capabilities, the clinical deployment of conditional diffusion models has been severely throttled by their computational latency. The iterative nature of the Markovian reverse process typically requires executing a massive neural network (such as a multi-million parameter U-Net) for hundreds or thousands of sequential timesteps[2]. Generating a dense, high-resolution 3D volume using standard DDPMs can take tens of minutes, rendering them entirely unsuitable for time-critical workflows, such as intraoperative surgical imaging or high-throughput emergency radiology[2]. Consequently, recent literature heavily prioritizes the architectural optimization of diffusion frameworks.

A. Adversarial-Diffusion Hybrid Architectures

To circumvent the sequential bottleneck, researchers are successfully hybridizing the stable distribution modeling of diffusion priors with the rapid inference speeds of adversarial networks[2]. Zhao et al. (2024) proposed DiffGAN, an adversarial diffusion model explicitly engineered to accelerate MRI reconstruction from highly undersampled k-space data[2]. In DiffGAN, the traditional denoising steps are governed by a Generative Adversarial Network (GAN) architecture[2]. By utilizing an adversarial loss function, the network is trained to take significantly larger "leap" steps over the diffusion trajectory, drastically reducing the total number of timesteps required for convergence[2].

Furthermore, DiffGAN replaces the computationally devastating global self-attention mechanisms of standard diffusion models with a Local Vision Transformer (LVT) generator[2]. The

LVT strictly bounds the attention mechanisms to localized spatial windows, vastly reducing the quadratic computational overhead while successfully preserving the high-frequency micro-textures essential for clinical evaluation[2]. Validated across the FastMRI, IXI, and MRNet datasets, DiffGAN achieved the stability of a diffusion model at the operational speed of a GAN, recording state-of-the-art PSNR and SSIM metrics while operating in a fraction of the traditional elapsed time[2].

A parallel methodology was adopted by Lu et al. (2026), who engineered the 3D Adversarial Diffusion model (3DAD) to facilitate the rapid synthesis of high-resolution, thin-slice CT volumes from highly compressed thick-slice archival data[47]. 3DAD leverages a deeply integrated generator-discriminator paradigm to enforce high-fidelity generation utilizing a highly reduced number of diffusion steps[47]. Validated rigorously in clinical trials for the detection of hepatocellular carcinomas and spinal compression fractures, 3DAD accurately hallucinated diverse, complex fracture morphologies that standard GANs smeared, maintaining diagnostic integrity while meeting the strict throughput requirements of high-volume radiological screening protocols[47].

B. Residual Shifting and Latent Optimization

Pushing the boundaries of ultra-low latency, Safari et al. (2025) developed Res-SRDiff, an highly efficient diffusion probabilistic model designed for real-time MRI super-resolution[45]. Res-SRDiff fundamentally alters the standard reverse process by incorporating a residual error shifting mechanism[45]. By continuously tracking and propagating the residual errors across the sampling trajectory, the architecture allows the model to safely bypass redundant denoising stages without accumulating catastrophic geometric drift[45]. Strikingly, Res-SRDiff achieved pristine, high-fidelity image reconstruction using a mere four sampling steps, plummeting the total computation time to under one second per slice[45]. When compared to conventional Denoising Diffusion Implicit Models (DDIMs), which typically demand 20 to 50 seconds per slice, Res-SRDiff represents a watershed achievement, enabling the viable integration of diffusion-based super-resolution directly onto the hardware consoles of clinical scanners[45].

An alternative strategy for acceleration focuses on optimizing the starting parameters of the reverse process. Bangun (2024) introduced the Regularized 3D Diffusion Model (R3DM), which synergizes advanced numerical optimization algorithms with 3D diffusion priors[50]. Rather than initiating the reverse process from pure Gaussian noise, R3DM rapidly traverses the latent space to establish an optimized, semi-structured initialization point[50]. This significantly reduces the total noise variance required to achieve high-quality MRI slice synthesis, drastically shortening the necessary denoising trajectory[50]. Similarly, Li et al. (2024) introduced McDiff (and DiffMSR), demonstrating that explicitly forcing the diffusion model to operate within a highly compact, low-dimensional latent space ensures that only a minimal number of simple iterations are required to generate accurate, high-frequency structural priors for multi-contrast MRI[52].

Table I. Summary of Advanced Algorithmic Efficiency Optimizations

Framework	Efficiency Mechanism	Computational Benefit	Primary Clinical Application
DiffGAN (Zhao et al., 2024)	Adversarial leap-stepping & Local Vision Transformers (LVT).	Avoids global attention scaling overhead; rapid inference.	Accelerated undersampled k-space MRI.
3DAD (Lu et al., 2026)	3D Adversarial Discriminator guidance.	Permits larger diffusion step sizes; fewer total iterations.	Thick-to-thin CT volume upsampling.
Res-SRDiff (Safari et al., 2025)	Residual error shifting propagation.	Convergence in 4 steps; <1 second latency per slice.	Real-time Prostate & Brain MRI analysis.
R3DM (Bangun, 2024)	Optimization-guide d latent initialization.	Reduced total denoising trajectory and noise variance.	3D Volumetric MRI synthesis.
McDiff (Li et al., 2024)	Highly compact low-dimensional latent space projection.	Minimizes iteration count required for high-frequency priors.	Multi-contrast MRI super-resolution.

VII. Comprehensive Discussion and Synthesis

A rigorous synthesis of the extensive literature spanning 2023 to 2026 reveals a profound evolution in the underlying philosophy of computational biomedical imaging. The integration of conditional diffusion models into the 3D reconstruction of tissues from sparse 2D scans is not merely an incremental upgrade in image quality; it represents a fundamental paradigm shift from *perceptual approximation* to *physically constrained generative modeling*.

The primary revelation is the critical maturation regarding data fidelity. Early deep learning models, particularly unconstrained GANs, were optimized aggressively for perceptual metrics (e.g., LPIPS or visual acuity)[3]. While these networks produced aesthetically pleasing scans, they were clinically dangerous, prone to erasing micro-lesions or hallucinating non-existent tumors in an effort to smooth the image distribution[11]. The deployment of mathematically rigid frameworks like Dual-Coupled PnP ADMM[28] and DOLCE[2] establishes a new standard: generative priors must be strictly, permanently subordinated to the physical measurement

operators of the imaging hardware. By ensuring that any high-frequency anatomical detail hallucinated by the diffusion model is tightly bounded by the actual raw k-space or sinogram measurements, these frameworks eliminate the risk of diagnostic hallucination, ensuring clinical safety[3].

Secondly, the literature demonstrates immense ingenuity in solving the "dimensionality conflict" of 3D medical imaging. The persistent reliance on 2D models due to memory constraints initially resulted in catastrophic inter-slice discontinuities[17]. However, algorithms such as ISCS[20], DF-DiffVSR[7], and TSSC-Net[24] have successfully circumvented this by treating the z-axis (depth) mathematically analogous to the temporal axis in video processing. The realization by Du et al. (2026) that physical continuity in anatomical space corresponds directly to geometric continuity within the latent Gaussian noise manifold is particularly profound[20]. Utilizing Spherical Linear Interpolation (Slerp) to synchronize the stochastic noise seeds proves that manipulating the foundational mathematical conditions of the reverse process is often vastly more efficient than attempting to brutally scale network architecture into 3D space[20].

Finally, the shift toward zero-shot generalization and retrieval augmentation indicates a critical move toward global, scalable deployment. Medical datasets are notoriously siloed, and the hardware ecosystems across different hospital networks are highly heterogeneous[2]. The innovations demonstrated by DACT—utilizing differentiable Sinkhorn Optimal Transport to dynamically map non-linear contrast shifts between low-field and high-field scanners purely based on marginal distributions[40]—and ReBrain—leveraging external database retrieval to guide cross-modal CT-to-MRI translation[37]—prove that models no longer require perfectly paired, massive local datasets to function effectively. These adaptable algorithms are uniquely positioned to be deployed in diverse clinical environments, automatically calibrating to the specific physics of local hardware without necessitating continuous, expensive retraining[41].

VIII. Conclusion

The 3D reconstruction of biomedical tissues from sparse, highly anisotropic 2D CT and MRI slices has been undeniably revolutionized by the integration of conditional diffusion models. By successfully overcoming the mathematical limitations of classical interpolation and the clinical unreliability of adversarial networks, diffusion models provide an unparalleled, robust capability to hallucinate anatomically faithful tissue boundaries and restore high-frequency diagnostic details from severely corrupted data.

The extensive literature reviewed herein—encompassing over 20 unique papers and frameworks including ISCS, BBDM/ISTA, DOLCE, Dual-Coupled PnP ADMM, ReBrain, DACT, DiffGAN, and Res-SRDiff—highlights comprehensive, elegant solutions to the inherent bottlenecks of generative medical AI. The severe challenge of inter-slice discontinuity has been decisively resolved through the implementation of latent noise correlation (Slerp), deformable optical flow fields, and advanced Mamba state-space modules. The critical mandate for diagnostic reliability is currently being met by fusing diffusion priors with explicit physical forward models, utilizing advanced numerical solvers to enforce strict measurement fidelity and eliminate generative hallucinations. Furthermore, the prohibitive latency of the reverse

diffusion process is being aggressively minimized through adversarial hybridization, residual shifting, and compact latent space projection, bringing inference times down to the sub-second thresholds required for real-time clinical workflows.
Future research must prioritize closing the hardware-software loop, optimizing memory-efficient fully 3D architectures, and expanding the robustness of zero-shot generalization techniques against extreme pathological anomalies. As these highly advanced algorithmic frameworks transition from theoretical literature to integration within the embedded systems of commercial radiological scanners, they hold the profound potential to drastically minimize patient radiation exposure, drastically reduce scan times, and fundamentally democratize global access to high-fidelity, isotropic 3D medical imaging.

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